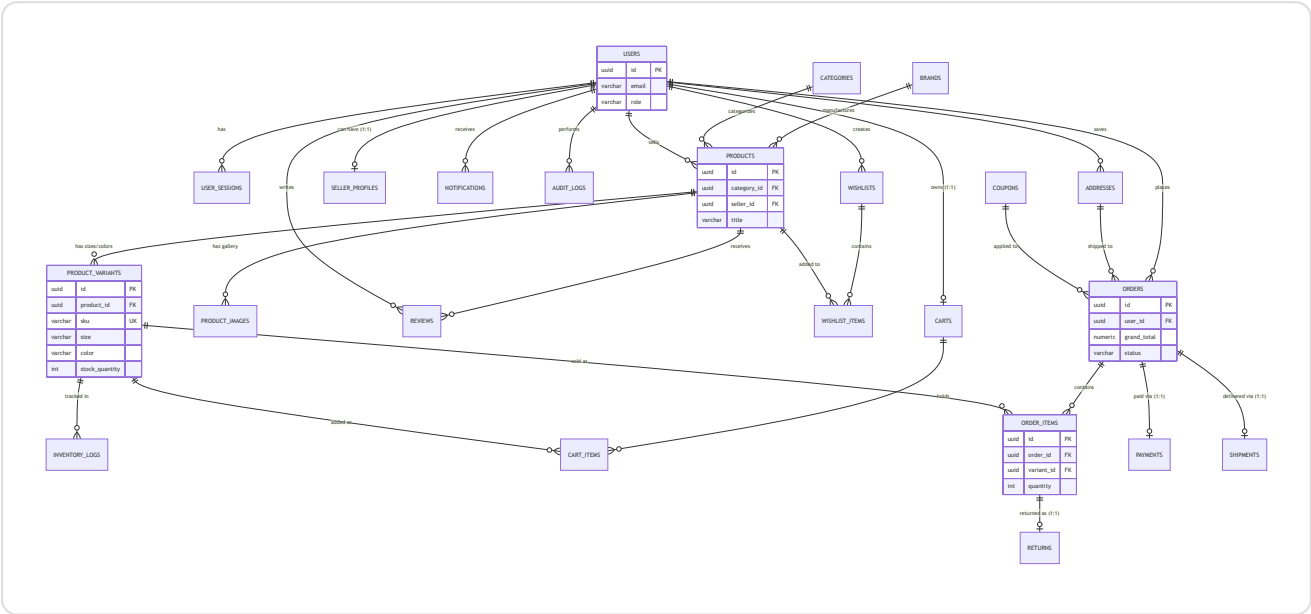


Advanced E-Commerce Architecture (PostgreSQL)

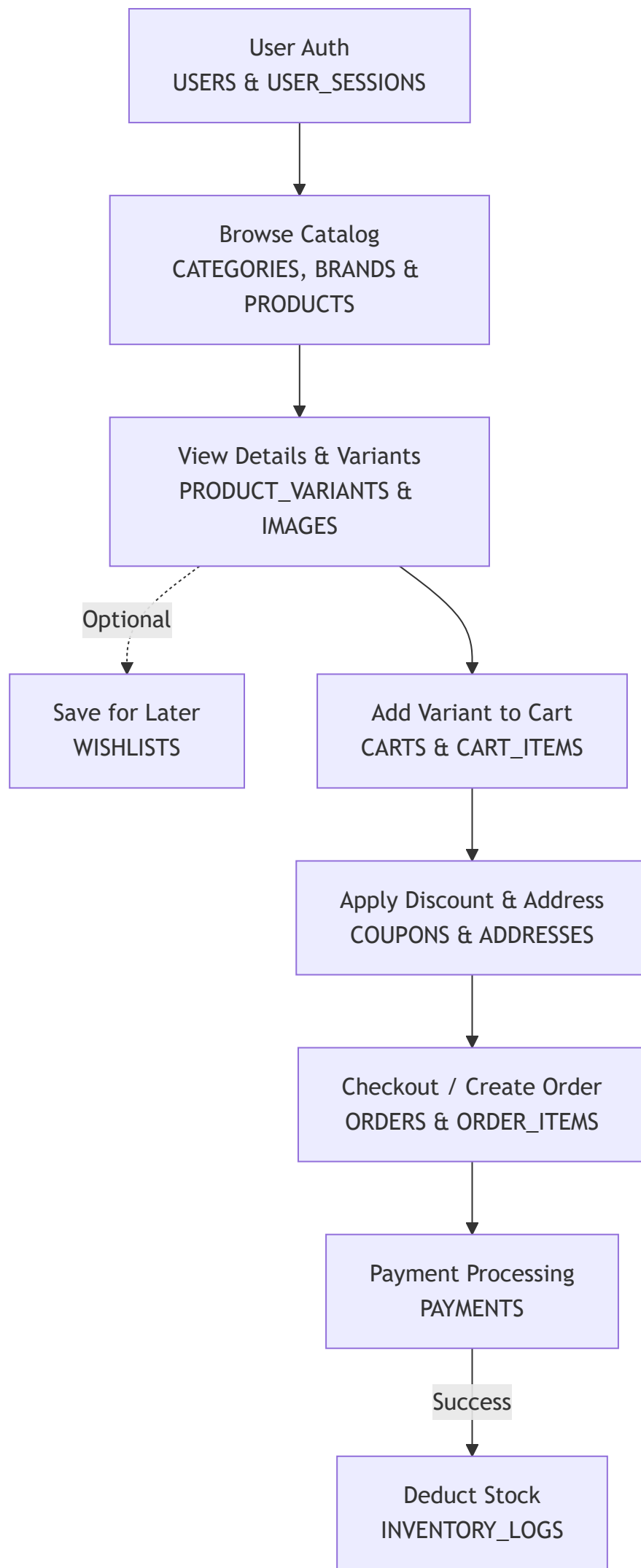
1. Entity Relationship Diagram (ERD)

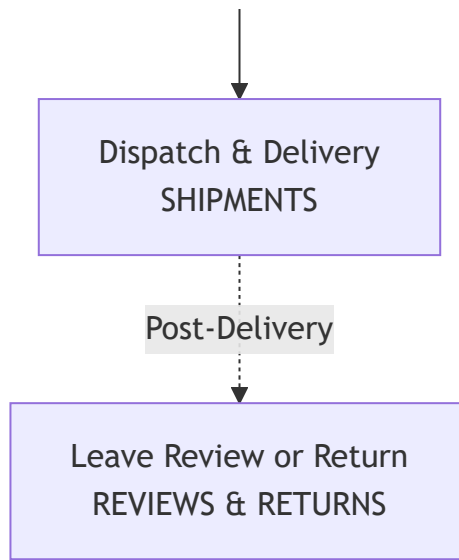
Below is the complete relational structure of the database, highlighting the Multi-Vendor and Apparel-Variant capabilities.



2. Flow of Understanding (How Data Moves)

This illustrates the sequential lifecycle of a user interacting with the apparel marketplace.





Step 1: Identity & Sellers

Data starts at **USERS**. Roles define access (Customer, Seller, Admin). Sellers manage their store details via **SELLER_PROFILES**. **USER_SESSIONS** secures the JWT lifecycle.

Step 2: Apparel Variations

PRODUCTS act as the parent container. The actual physical items exist in **PRODUCT_VARIANTS** (handling specific Sizes, Colors, and SKUs). **PRODUCT_IMAGES** supports multiple angles.

Step 3: Cart & Engagement

Users add specific **PRODUCT_VARIANTS** to **CART_ITEMS**. They can also save items to **WISHLISTS**. Stock changes are securely audited in **INVENTORY_LOGS**.

Step 4: Full Order Lifecycle

Checkout shifts data to **ORDERS** (applying **COUPONS**). **PAYMENTS** verifies funds, **SHIPMENTS** tracks courier details, and **RETURNS** handles post-sale refunds.